KS

KETUL SHAH

IT SUPPORT | SYSTEMS ADMINISTRATION LAB

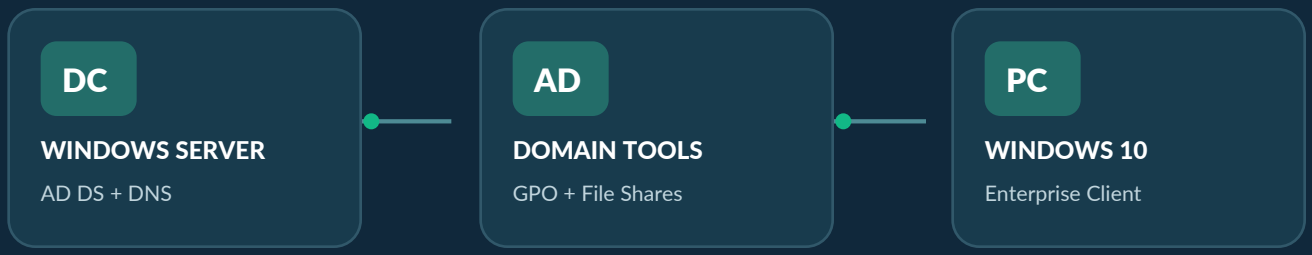
HANDS-ON PROJECT PORTFOLIO

IT SUPPORT HOME LAB

Windows Server 2022, Active Directory, Group Policy, file services and endpoint security.

LAB ARCHITECTURE

Virtualized client-server environment



01	02	03	04
ACTIVE DIRECTORY	GROUP POLICY	FILE SERVICES	SECURITY

PERSONAL TECHNICAL LAB / 2026

CONFIGURATION • TESTING • DOCUMENTATION

AT A GLANCE

Four practical lab projects.

A guided, hands-on Windows administration lab focused on core entry-level help desk and IT support workflows. Screenshots show configuration steps and selected client-side outcomes.

01 ACTIVE DIRECTORY — P. 03–05
Windows Server, AD DS, DNS, OUs, users and groups

02 GROUP POLICY — P. 06–08
Password policy, desktop settings, restrictions and testing

03 FILE SERVICES — P. 09–13
Share/NTFS access, drive mapping, quotas and file screening

04 SECURITY POLICIES — P. 14–15
Account lockout testing and user rights assignment

HOW TO READ THIS PORTFOLIO

Configuration screenshots demonstrate settings; client screenshots demonstrate observed behaviour. Where a specific test or final policy state is not visible, the text says so.

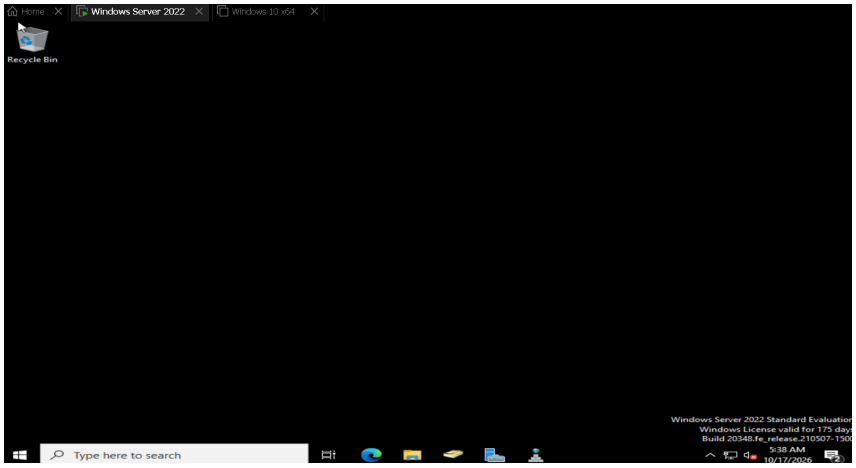
FIELD NOTES / 01 / ACTIVE DIRECTORY

Deploying Windows Server & AD DS

Built a Windows Server 2022 Standard Evaluation virtual machine in VMware Workstation and installed the Active Directory Domain Services (AD DS) and DNS Server roles.

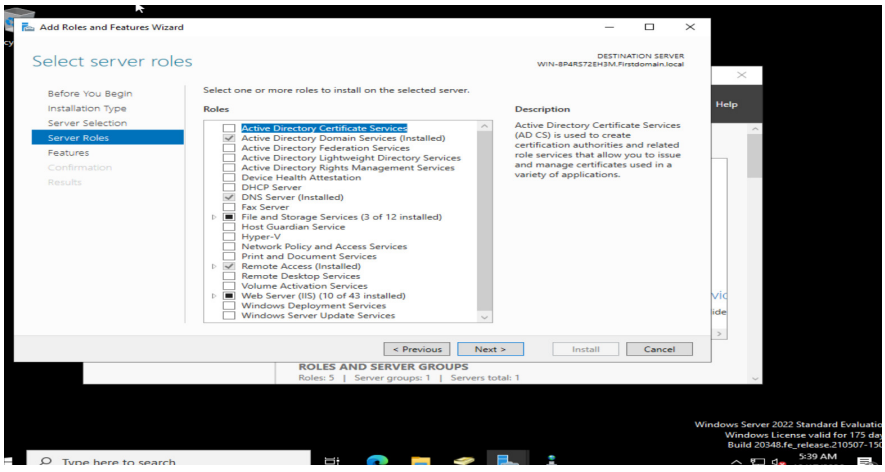
VMWARE | SERVER ROLES | AD DS + DNS

01



Windows Server 2022 virtual machine running — VMware Workstation

02



AD DS and DNS Server shown as installed — Server Manager

RESULT A functioning server VM and the required directory-service roles are visible.

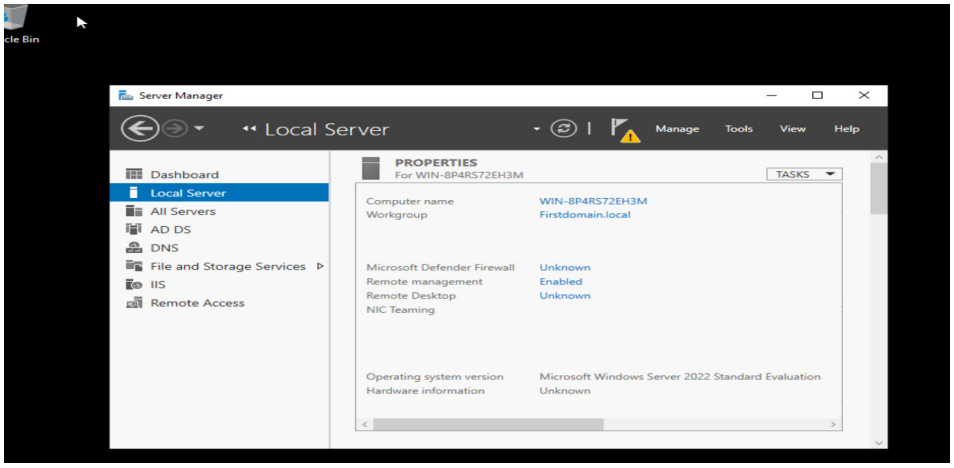
FIELD NOTES / 01 / ACTIVE DIRECTORY

Domain Services & DNS

Reviewed the server configuration for Firstdomain.local and inspected Active Directory-integrated DNS zones used in domain name resolution.

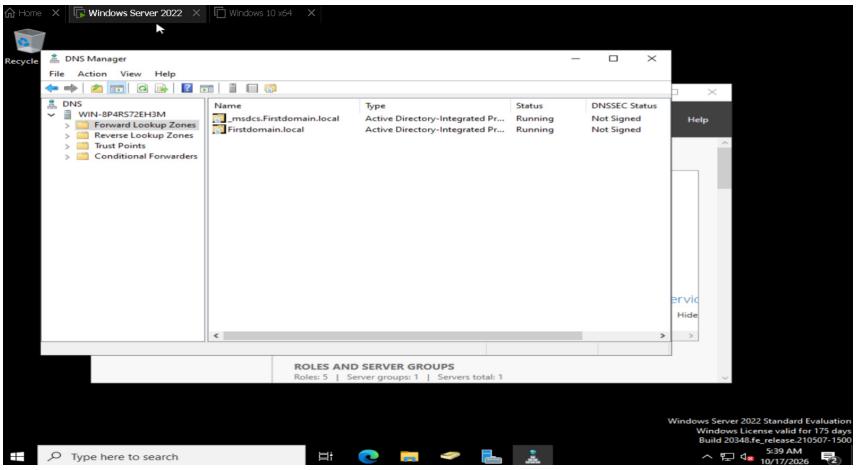
FIRSTDOMAIN.LOCAL | SERVER MANAGER | DNS MANAGER

03



Local Server operating-system and host properties — Windows Server 2022

04



DNS zones for Firstdomain.local and _msdcs — AD-integrated zones

EVIDENCE The screenshots show the DNS infrastructure; no health-check output is provided.

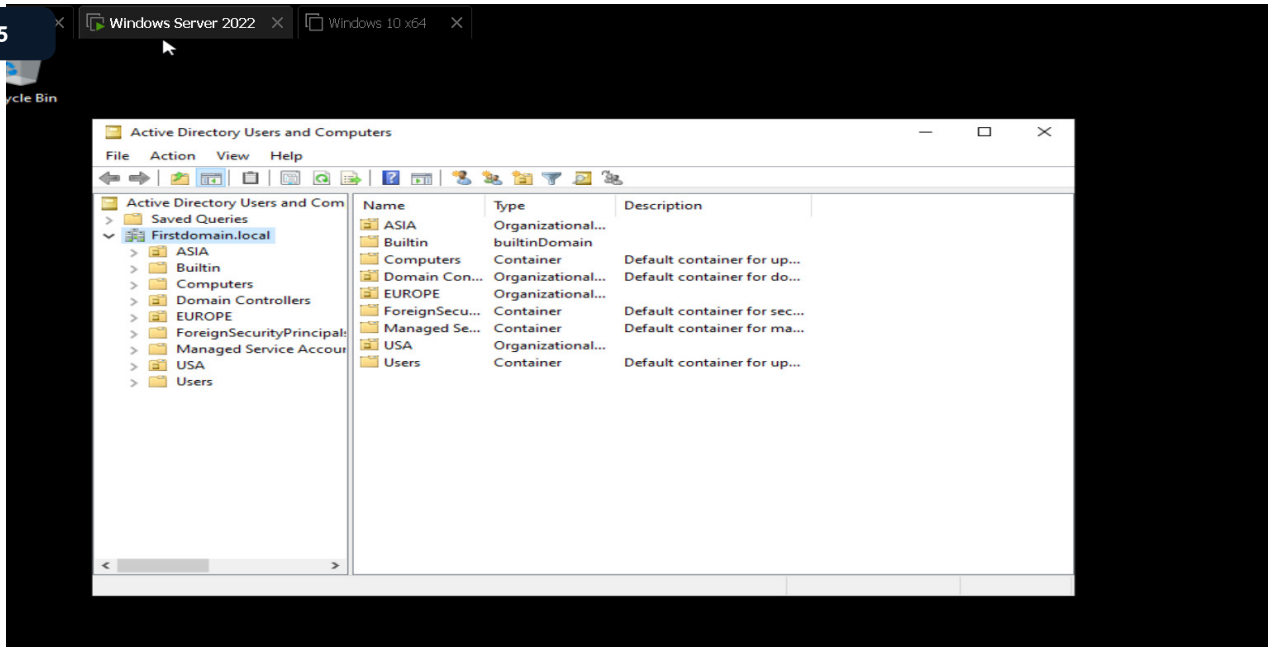
FIELD NOTES / 01 / ACTIVE DIRECTORY

Organizational Units & Identity

Created regional Organizational Units and practiced the Active Directory workflow for arranging user accounts and groups. Reviewed Security and Distribution group purposes.

ACTIVE DIRECTORY USERS & COMPUTERS | OU STRUCTURE

05



Firstdomain.local with ASIA, EUROPE and USA OUs — ADUC console

OU STRUCTURE Created ASIA, EUROPE and USA beneath Firstdomain.local.

IDENTITY MANAGEMENT Practiced user-account and group creation within OUs.

GROUP TYPES Security groups grant permissions; Distribution groups support email lists.

SCOPE Individual user objects and group memberships are not shown in this view.

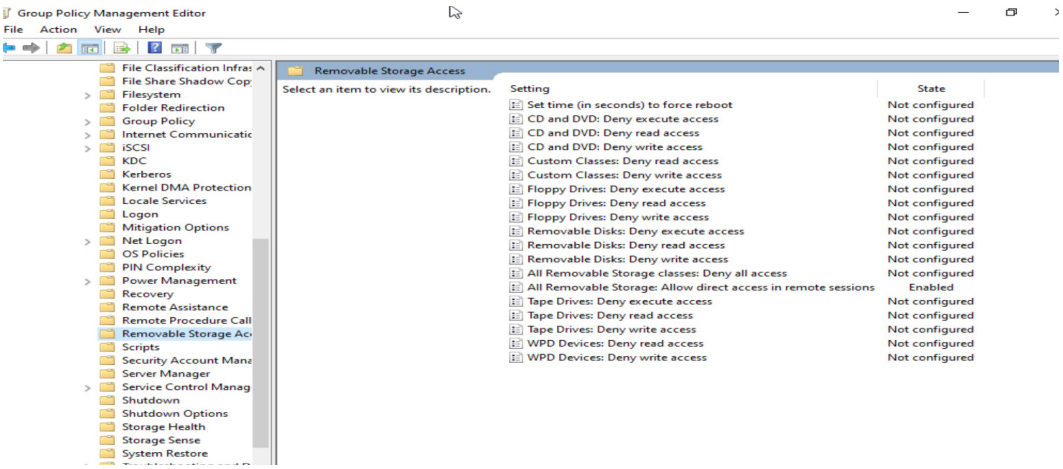
FIELD NOTES / 02 / GROUP POLICY

Password Rules & Desktop Standards

Used Group Policy Management Editor to work with Computer and User Configuration and define central password and desktop settings.

10-CHAR MINIMUM | COMPLEXITY ON | DESKTOP WALLPAPER

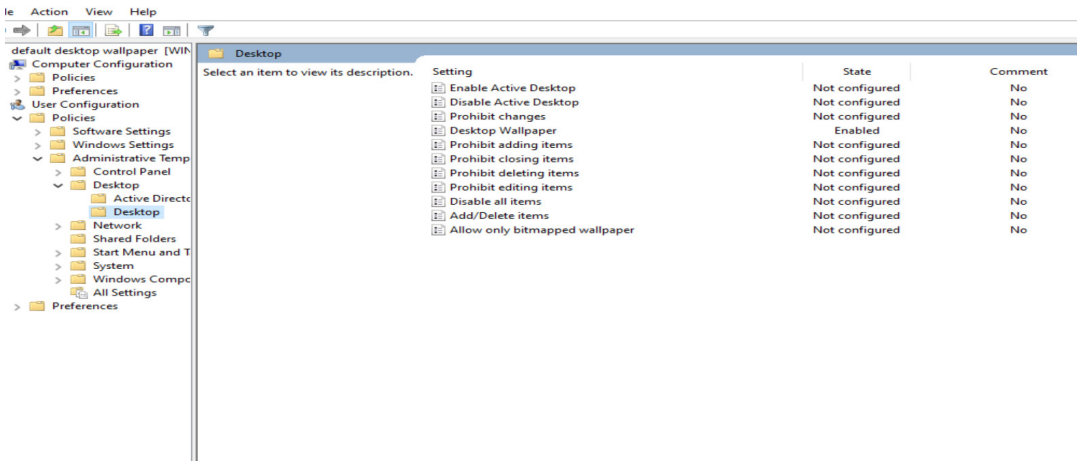
06



Password Policy GPO with configured security values

See details below

07



Desktop Wallpaper policy set to Enabled

User Configuration

SETTINGS 10 characters; maximum 180 days; minimum age 30 days; complexity enabled.

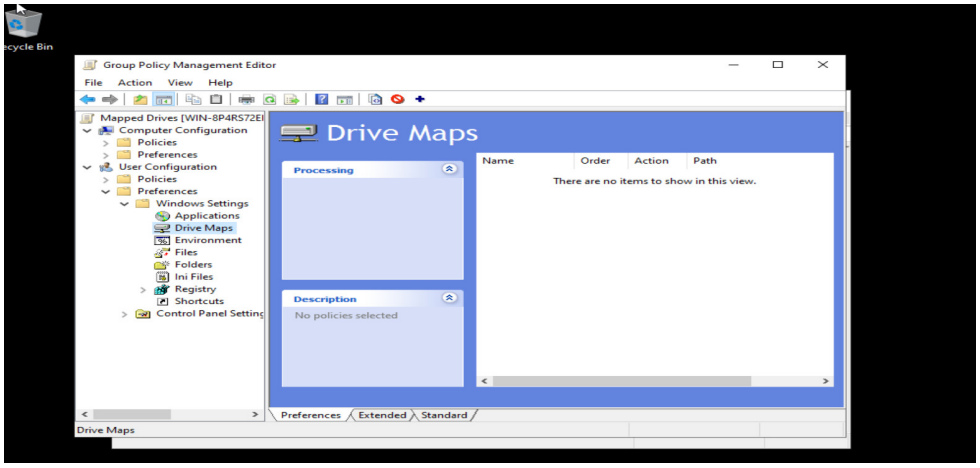
FIELD NOTES / 02 / GROUP POLICY

Restrictions & Targeted Deployment

Configured a policy to block Control Panel and PC settings. Linked user and computer-oriented GPOs to Organizational Units under USA and reviewed GPO inheritance.

CONTROL PANEL | OU LINKING | GROUP POLICY MANAGEMENT

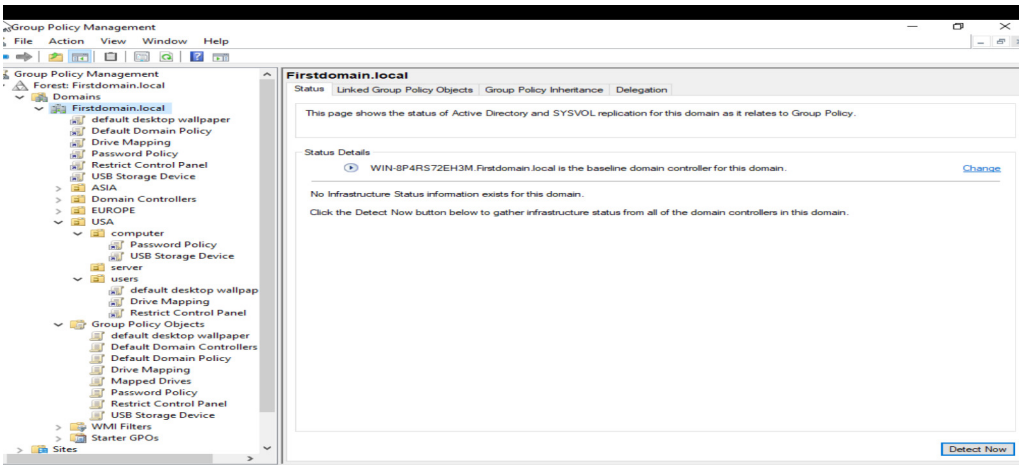
08



Control Panel restriction set to Enabled

User policy

09



Domain and USA OU GPO link structure

GPMC console

SCOPE GPOs also linked at the domain level may affect users outside USA.

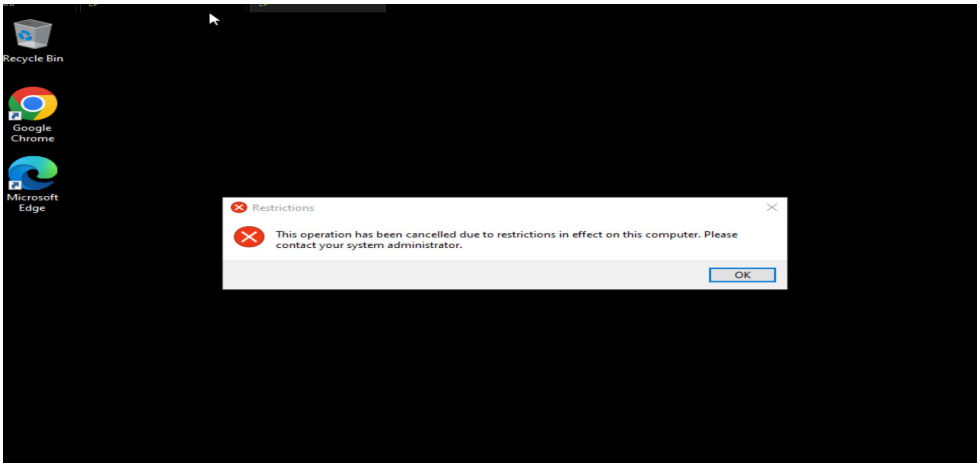
FIELD NOTES / 02 / GROUP POLICY

Windows Client Test & USB Controls

Set up a Windows 10 Enterprise client for domain-lab testing. A Windows restriction message was observed after Group Policy work; removable-storage policies were also explored.

WINDOWS 10 ENTERPRISE | CLIENT TEST | REMOVABLE STORAGE

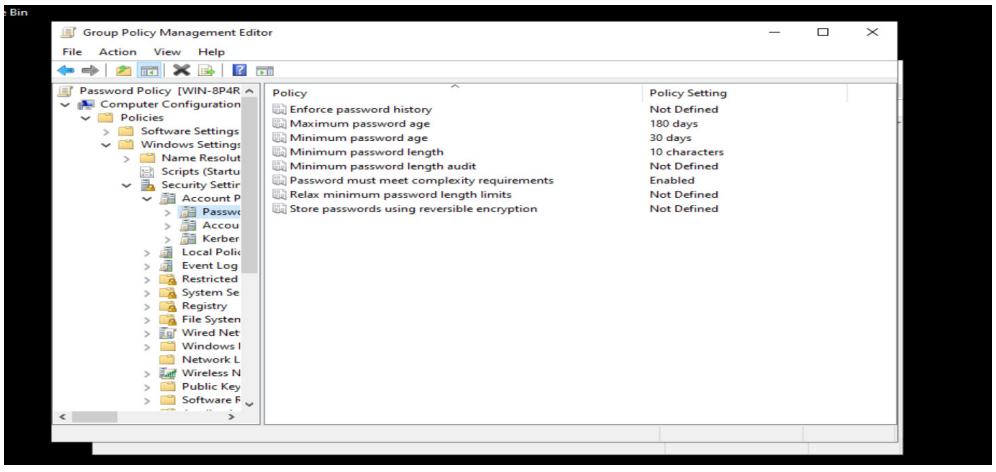
10



Client reports an administrative restriction

Observed client result

11



Removable Storage Access policy editor

Settings inspection

VERIFICATION Shown USB-deny settings are Not Configured; USB blocking is not proven.

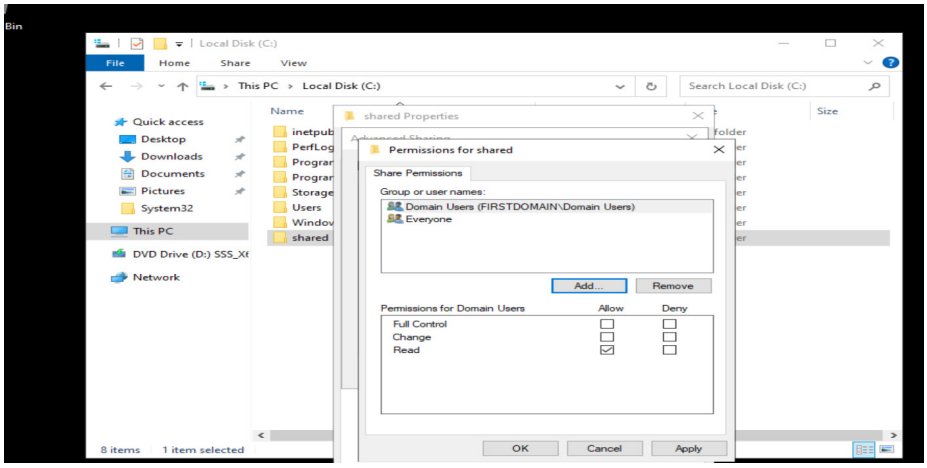
FIELD NOTES / 03 / FILE SERVICES

Shared Folders & Access Control

Created C:\shared on Windows Server 2022, configured network sharing, and reviewed the interaction between share permissions and NTFS file-system permissions.

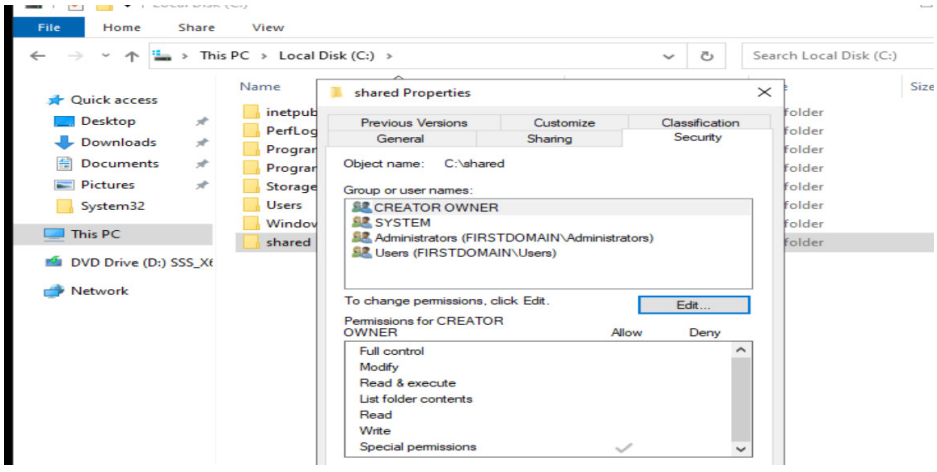
SMB SHARES | SHARE PERMISSIONS | NTFS

12



Domain Users assigned Read share permission — C:\shared

13



NTFS Security tab and permission categories — File-system ACLs

ACCESS Effective network access is constrained by both share and NTFS permissions.

FIELD NOTES / 03 / FILE SERVICES

Group Policy Drive Mapping

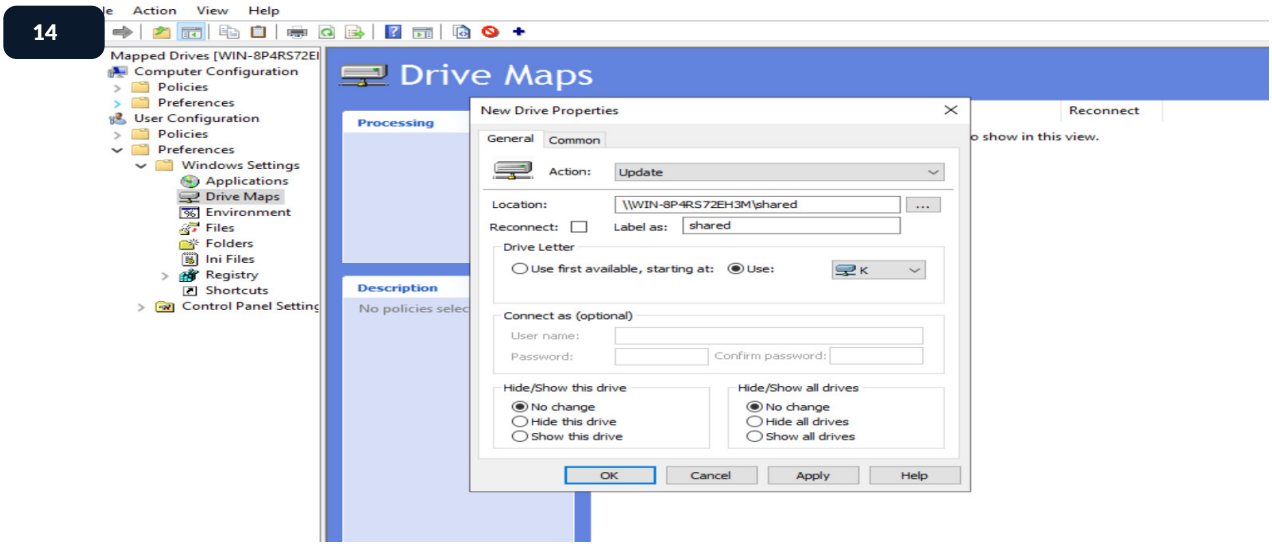
Built a mapped-drive preference using the Update action, a UNC share path and a designated drive letter. This lets IT administrators deploy network drives centrally.

ACTION / UPDATE

LETTER / K:

SHARE / C:\shared

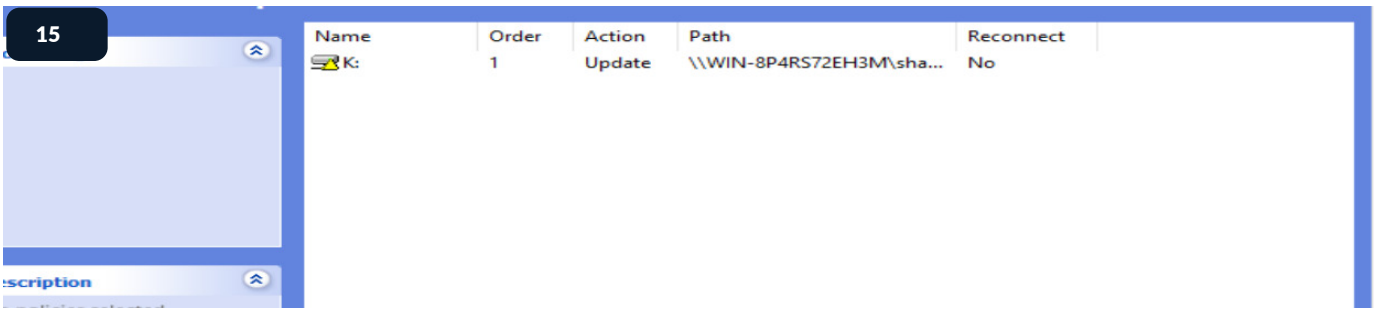
14



Drive map properties: Update action to K:

Group Policy Preferences

15



K: drive preference visible in the GPO list

Mapped Drives

NETWORK SHARE

Update • Drive K:

`\\WIN-8P4RS72EH3M\shared`

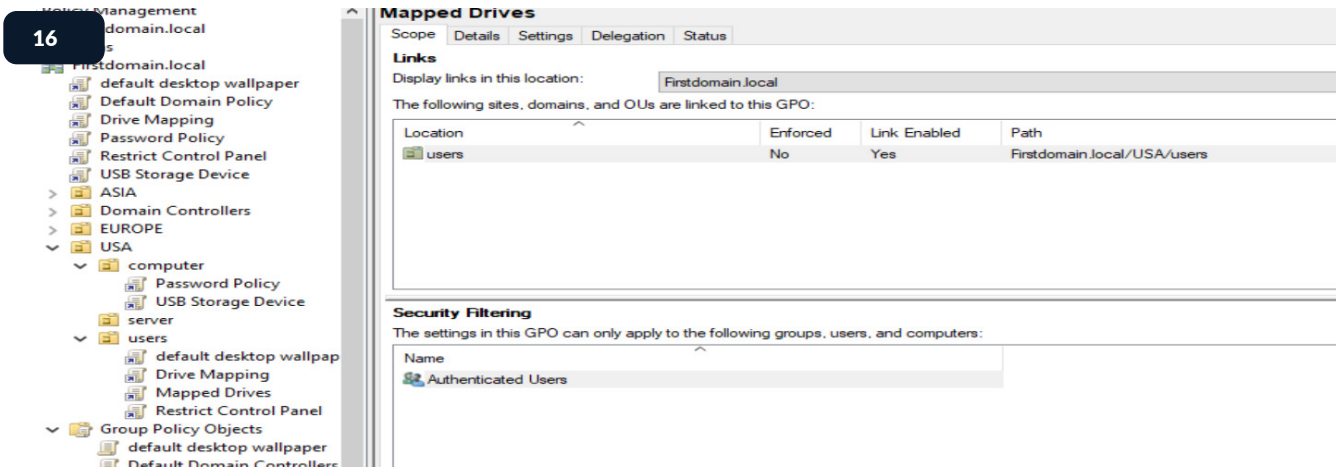
FIELD NOTES / 03 / FILE SERVICES

Deploying Drives to Domain Users

Linked the Mapped Drives GPO to Firstdomain.local/USA/users and used a domain-signed-in Windows 10 client to view mapped network locations.

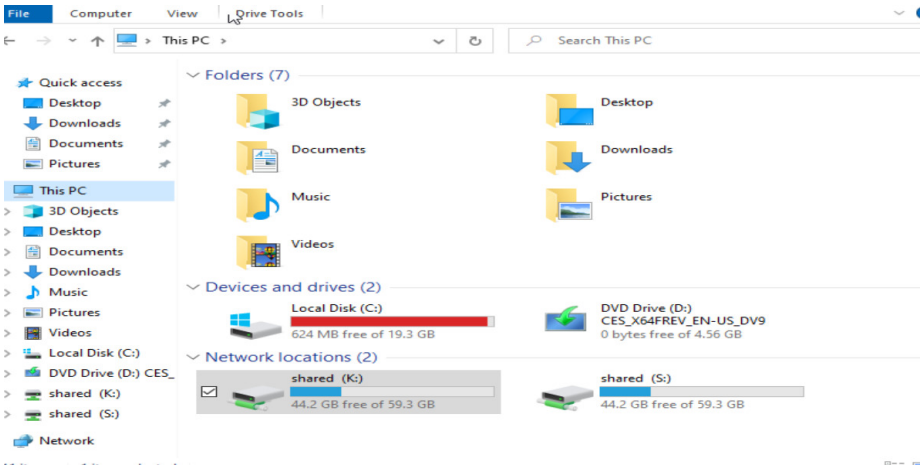
USA / USERS OU | AUTHENTICATED USERS | K: + S: VISIBLE

16



Mapped Drives policy linked to USA/users OU — GPO Scope

17



Network drives K: and S: in File Explorer — Client verification

RESULT Both drive letters appear; persistence after restart is not separately evidenced.

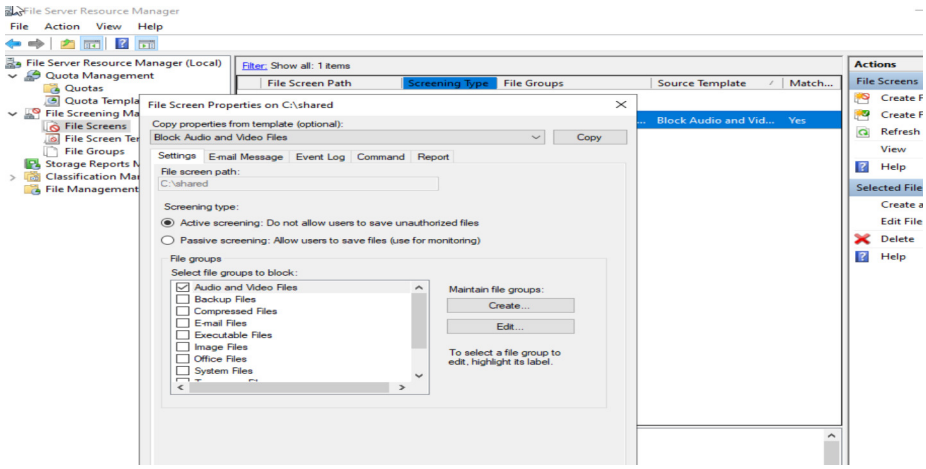
FIELD NOTES / 03 / FILE SERVICES

File Screening & Storage Quotas

Configured File Server Resource Manager policies for the shared folder: an active audio/video file screen and a hard quota to limit stored data.

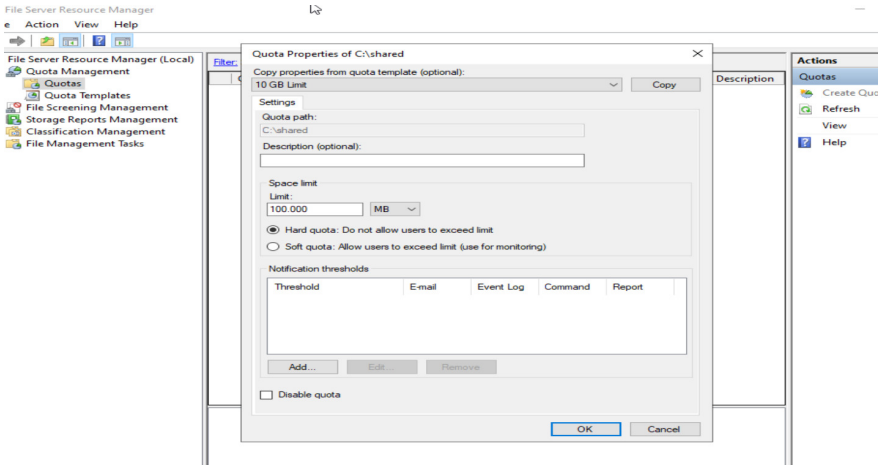
FSRM | ACTIVE FILE SCREEN | 100 MB HARD QUOTA

18



Active screening for audio/video file group C:\shared

19



Hard storage quota set to 100 MB C:\shared

VERIFICATION Configured settings are shown; block/limit enforcement test is not captured.

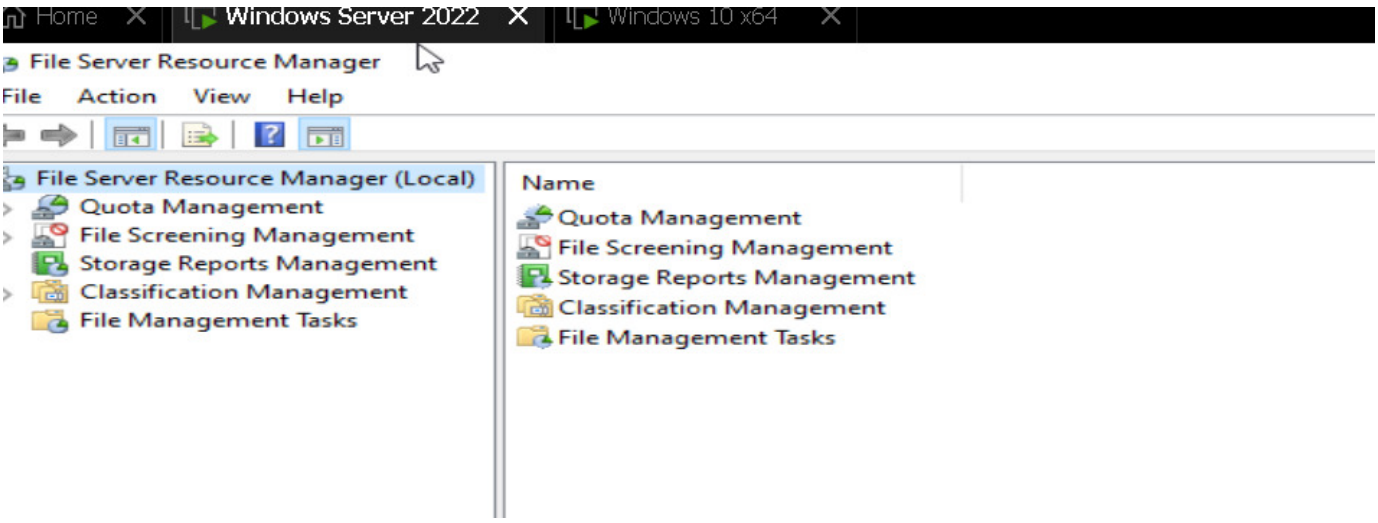
FIELD NOTES / 03 / FILE SERVICES

File Server Resource Manager Tools

Explored the File Server Resource Manager console and the management areas available for file-server storage governance.

FSRM CONSOLE | QUOTA MANAGEMENT | FILE SCREENING

20



FSRM console available on Windows Server Server tools

QUOTA MANAGEMENT Supports hard/soft storage limits and threshold notifications.

FILE SCREENING Can block or monitor files in configured file groups.

OTHER FEATURES Storage reports, classification management and file management tasks.

CONFIGURED The current lab uses a 100 MB hard quota and active audio/video screening.

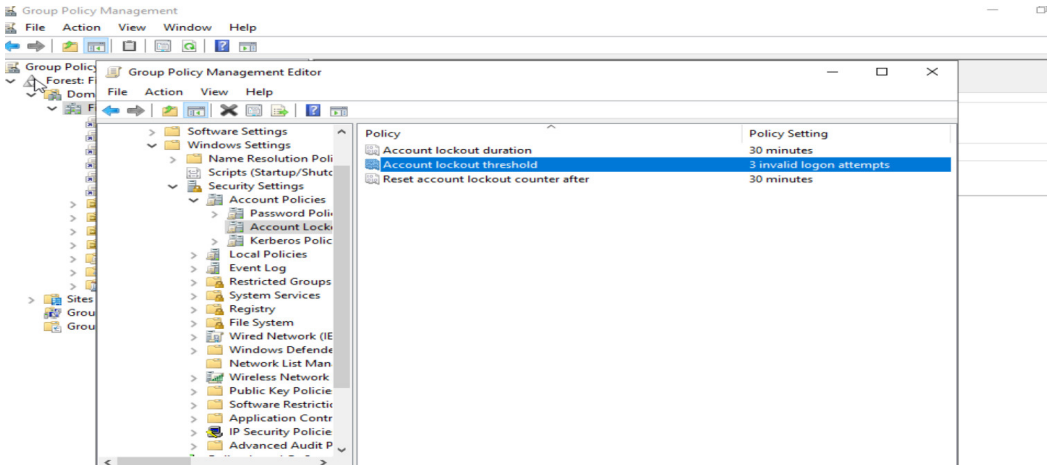
FIELD NOTES / 04 / SECURITY POLICIES

Account Lockout Configuration & Test

Configured a lockout policy intended to slow repeated password attempts, then observed a domain-user account lockout at the Windows 10 login screen.

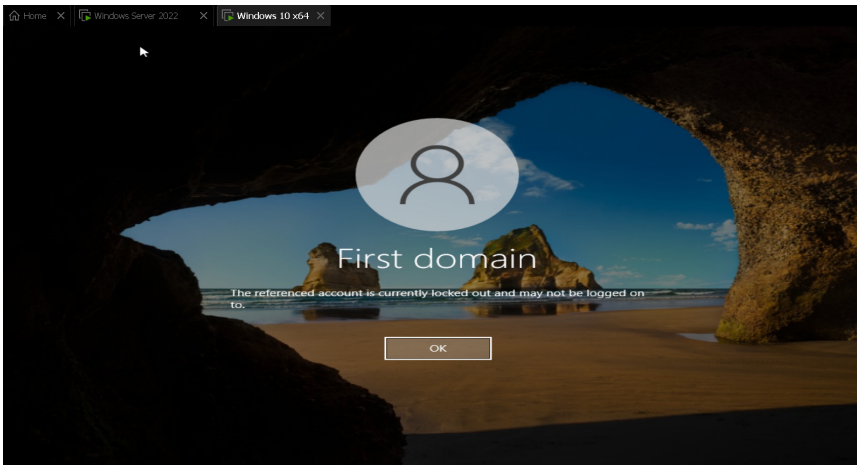
3 INVALID ATTEMPTS | 30-MIN LOCKOUT | 30-MIN RESET

21



Threshold 3; lockout and reset after 30 minutes — GPO editor

22



Domain login blocked by account lockout — Observed result

OUTCOME Windows 10 displays the expected locked-account message.

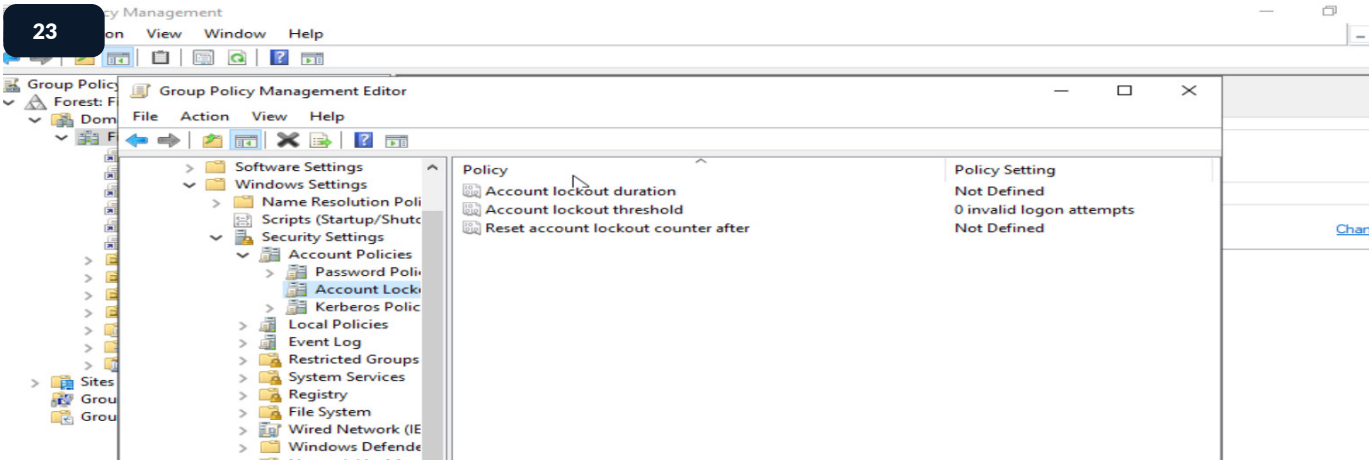
FIELD NOTES / 04 / SECURITY POLICIES

Policy Exploration & User Rights

Compared account-lockout behaviour when a threshold is set to zero, then created a User rights GPO and reviewed the available User Rights Assignment settings.

ACCOUNT POLICIES | USER RIGHTS | LEAST PRIVILEGE

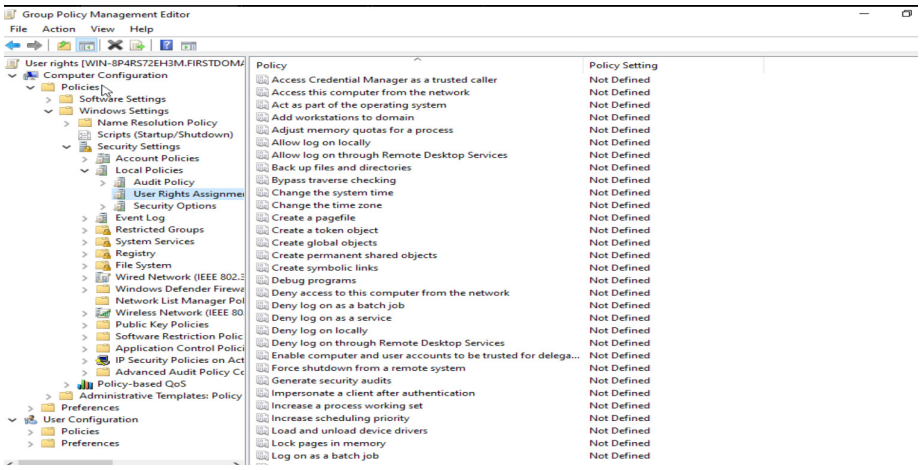
23



Lockout threshold 0 in a policy editor

Disables threshold lockout

24



User rights GPO: available rights assignments

Not Defined in this view

SCOPE The image does not show rights granted or denied; this was policy exploration.

SUMMARY

What this home lab demonstrates

Technical projects document a practical Windows domain environment and a repeatable approach to administering, deploying and checking configuration changes.

01

IDENTITY & DIRECTORY

Active Directory Domain Services, OUs, user/group administration, DNS

02

CENTRAL MANAGEMENT

GPO configuration, OU targeting, user and computer policy, mapped drives

03

FILE SERVICES

SMB shares, NTFS/share permission concepts, FSRM quotas and screening

04

SECURITY & VALIDATION

Account lockout configuration, Windows client test, rights review

EVIDENCE & LIMITATIONS

Screenshots show actual lab settings and selected outcomes. USB blocking, exact User Rights enforcement, and file-screen/quota rejection tests are not shown. Group-policy results can depend on scope, inheritance, filtering and refresh.

LEARNING REFERENCE

Tutorial-guided Windows Server home lab ↗

Built and documented by Ketul Shah